HOMEWORK 2
BIOSTATISTICS 755
DUE FEBRUARY 12, 2026

1. **Insulin study (36 rabbits; 3 treatment groups).**
The longitudinal data from an insulin study include 36 rabbits. Twelve rabbits were randomly assigned to each of three insulin mixtures:

    - **Group 1:** standard insulin mixture
    - **Group 2:** mixture with **1% less protamine** than standard
    - **Group 3:** mixture with **5% less protamine** than standard

    Rabbits were injected at time 0, and blood sugar was measured at **7 time points**: $0, 0.5, 1.0, 1.5, 2.0, 2.5$, and 3.0 hours post-injection.

    The data file *"insulin"* is on the course website. The variables in the wide-format file are:

    - Column 1: rabbit ID
    - Column 2: insulin group
    - Columns 3–9: blood sugar at the 7 time points (in the order listed above)

    (a) **(10 points)** Create a spaghetti plot with separate panels for each group. Comment briefly on heterogeneity (between-rabbit differences in baseline level and/or time trends).

    (b) **(10 points)** Create a plot of the **group mean blood sugar over time** on a single figure (use different colors if submitting in color; otherwise use different line types).
    Comment on what mean time trend(s) seem plausible (e.g., profile/categorical time, linear, quadratic).

    (c) **(10 points)** Fit a **full interaction profile model** (i.e., treat time as categorical and include group, time, and group$\times$time). Fit the model under each covariance structure below.
    For each structure, submit only: (i) the estimated covariance matrix (or covariance parameter estimates, as appropriate) and (ii) AIC and BIC.

    i. Unstructured (UN)
    ii. Compound symmetry (CS)
    iii. Heterogeneous compound symmetry (CSH)

    (d) **(10 points)** Using **only** the estimated covariance/correlation from the **UN** fit, which structure (UN, CS, or CSH) seems most reasonable? Briefly justify your choice.
    Do not use AIC/BIC for this question.

(e) **(10 points)** Based on AIC and BIC from part (c), which covariance structure provides the best fit? State your answer and report the AIC/BIC values you used.

(f) **(10 points)** Perform a likelihood ratio test (LRT) comparing:

   i. UN vs. CSH

   Report the test statistic, degrees of freedom, and p-value. Do the LRT conclusions align with AIC/BIC?

(g) **(10 points)** Using the best-fitting covariance structure from part (c), test whether the **mean profiles over time differ by group** (i.e., test the group$\times$time interaction). State the null hypothesis and report the test result.

(h) **(10 points)** Does **time** have a statistically significant effect on blood sugar? (You may need to fit a reduced model depending on how you frame the test.) State the null hypothesis and report the test result.

(i) **(10 points)** Using the best-fitting covariance structure from part (c), estimate the **within-group change** from baseline (0 hr) to 0.5 hr in **Group 1**. Provide the estimate (and a CI if available) and interpret it in context.

(j) **(10 points)** Using the best-fitting covariance structure from part (c), interpret at least **two** model parameters in context. At least **one** interpretation must involve a **group$\times$time interaction** term (i.e., a difference in changes over time).